Conditional Reduction to General Relativity from Zero-Sum Constraints: Lovelock Uniqueness and the Group-Theoretic Origin of Dimension

Abstract

This work establishes a conditional reduction chain from a discrete ontological substrate to general relativity. The method does not assemble the Einstein field equations piece by piece; instead, it compresses the field equations into three structural properties—locality (L), second-order (O), and conserved source (C')—and then invokes the Lovelock uniqueness theorem to force them into Einstein form. The algebraic skeleton: zero-sum constraints fix codimension 1; the incidence matrix of a connected graph spans the zero-sum hyperplane and gives integer reachability; discrete Dirichlet energy is positive semidefinite with kernel the constants; continuum approximation combined with metric-volume consistency recovers the spatial metric; reversible/irreversible splitting directly gives signature $(1, m-1)$. The metric construction: closed-walk counting gives a parameter-free edge weight $w_{ij} \propto \phi_i \phi_j A_{ij}$; finite truncation $k = n$ restores locality. The dimension account is the weakest link: the core postulates do not constrain dimension (proposition: for every $m \geq 2$ there exists a model satisfying the postulates); $D = 4$ follows from two independent routes—physically, "propagating gravitons exist" requires the unique positive solution of $\frac{D(D-3)}{2} = 2$, and geometrically, the closure–motion correspondence principle $\dim \ker B = \dim \operatorname{im} B$ combined with edge-transitivity uniquely selects K_4. The premises of both routes are not consequences of the core postulates; dimension is therefore a conditional input.

Keywords: Lovelock theorem; general relativity; zero-sum constraint; discrete geometry; Hodge duality; dimension problem

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1 Introduction

The spacetime dimension $D = 4$ is an input rather than a conclusion within the standard physical framework [1,2]. Our approach starts from a discrete substrate containing no manifold, coordinates, metric, probability, or action, such that the shape of the Einstein field equations becomes a consequence derivable from the Lovelock uniqueness theorem.

Strategy. We do not assemble GR piece by piece. GR is compressed into three structural properties: (L) locality, (O) second-order, (C') conserved source. The substrate need only provide these three; the shape of the geometric field equations is then locked by the Lovelock theorem [3].

Scope. The Lovelock theorem's literal statement is "in 4 dimensions." This lever therefore cannot be used to select $D = 4$; dimension must be argued separately (§6).

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2 Core Postulates of the Substrate

The substrate declares only the following; no manifold, coordinates, dimension, distance, curvature, action, or probability weights appear:

- **State**: codimension-1 affine constraint on integer lattice (zero-sum);
- **Transport**: connected, reversible compensatory moves;
- **Energy**: Dirichlet-type quadratic form;
- **Time**: step index τ and terminal;
- **Record**: multiplicity multiset of closed classes;
- **Weights**: integer multiplicities.

The substrate axiom is $Z_0 = (\{\pm 1\}, \iota) + (\mathcal{W}_n, \mu)$, where ι is the direction involution and μ the uniform counting measure [4].

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3 Algebraic Skeleton

3.1 Zero-Sum Codimension and Transport Rank

Lemma 3.1 (Codimension 1). The zero-sum constraint $\sum_v x_v = 0$ gives only one linear condition:

$$\dim H_Q = m - 1, \qquad \{e_j - e_1\}_{j=2}^{m} \text{ is a basis.}$$

Corollary 3.2. To obtain "four dimensions," one must first have $m - 1 = 4$ or $m = 4$; neither is a consequence of the substrate postulates.

Lemma 3.3 (Transport rank and integer reachability).

$$\operatorname{im} B = H_Q, \qquad \operatorname{rank} B = m - 1,$$

$$x \in H_Q \cap \mathbb{Z}^C \implies \exists f \in \mathbb{Z}^E,\ Bf = x.$$

Proof. B is totally unimodular [5]; integer reachability follows from standard integer linear programming results. ■

3.2 Discrete Dirichlet Energy

Lemma 3.4. The discrete Dirichlet energy with edge weights K_e

$$E(f) = \sum_e K_e (f_i - f_j)^2 \geq 0, \qquad E(f) = 0 \iff f \text{ is constant,}$$

i.e. $E = f^{\mathsf{T}} L f$, $\ker L = \operatorname{span}\{\mathbf{1}\}$, $L(H_Q) \subseteq H_Q$, $L|_{H_Q} \succ 0$.

Lemma 3.5 (No preference ⇒ uniform conductance). If Γ is edge-transitive, then $K_e \equiv \kappa$.

Remark. This is conditional: path graphs P_4, P_5 have 2 edge orbits, and no-preference does not yield uniform conductance.

3.3 Local Quadratic Form and Spatial Metric

Lemma 3.6 (Metric-volume consistency inversion). Continuum approximation gives $E \to \int Q^{ij}\partial_i f\,\partial_j f\,\omega$. Requiring metric-volume consistency

$$\mu = \mathrm{vol}_h = \sqrt{\det h}\,\omega, \tag{3.1}$$

one obtains $Q = \sqrt{\det h}\,h^{-1}$, $\det Q = (\det h)^{d/2-1}$ (d = spatial dimension). For $d \neq 2$ the unique inverse is

$$h = (\det Q)^{\frac{1}{d-2}}\,Q^{-1}. \tag{3.2}$$

For $d = 2$, $\det Q = 1$, and $h \mapsto \Omega^2 h$ leaves Q invariant; only the conformal class remains.

Remark. (3.1) is the sole additional input in this step; it binds the Dirichlet quadratic form one-to-one to the spatial metric and removes conformal ambiguity for $d \neq 2$. The unique inverse holds for all $d \neq 2$ and does not select any dimension.

3.4 Splitting, Time, and Signature

Lemma 3.7 (Reversible/irreversible splitting).

$$T \cong H_Q \oplus \mathbb{R}_\tau \cong \mathbb{R}^m.$$

The regular bilinear form can only be $g = d\tau^2 - h$ or $g = -d\tau^2 + h$. Taking the former gives $\mathrm{signature}(g) = (1, m-1)$.

Mechanism. The substrate's compensatory moves are reversible (each edge has an inverse move); step progression is irreversible. Spatial directions carry the inverse structure; the step direction does not. The signature originates here—there is no need to single out a time direction among the spatial directions.

Boundary. $1 \oplus (m-1)$ and $h > 0$ are derived; which side carries the minus sign is a sign convention.

3.5 Curvature and Two-Derivative Uniqueness

Lemma 3.8. Given a smooth manifold M and nondegenerate g, the connection $\Gamma^a{}_{bc}$, curvature $R^a{}_{bcd}$, Ricci R_{ab}, and Einstein tensor G_{ab} are uniquely determined by standard differential geometry [6].

Lemma 3.9 (Uniqueness within the two-derivative family). Let $E_{ab} = \alpha R_{ab} + \beta R g_{ab} + \gamma g_{ab}$. Using $\nabla^a g_{ab} = 0$ and the contracted Bianchi identity $\nabla^a R_{ab} = \frac{1}{2}\nabla_b R$:

$$\nabla^a E_{ab} = \Big(\frac{\alpha}{2} + \beta\Big)\nabla_b R. \tag{3.3}$$

Thus when $\nabla R \not\equiv 0$, divergence-free $\iff$ $\beta = -\alpha/2$, i.e. $E_{ab} \in \mathrm{span}\{g_{ab}, G_{ab}\}$.

3.6 Lovelock Uniqueness and the GR Criterion

Lemma 3.10 (Lovelock uniqueness) [3]. In 4 dimensions, a symmetric second-rank tensor constructed from g and its derivatives up to second order, linear in the highest derivative, and satisfying $\nabla^a E_{ab} = 0$ must be

$$E_{ab} = c_1 G_{ab} + c_2 g_{ab}.$$

(The Gauss–Bonnet term is a topological term in 4 dimensions and does not enter the field equations.)

Corollary 3.11 (GR criterion).

$$(L) \wedge (O) \wedge (C') \implies G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}. \tag{3.4}$$

Here (C') requires that T_{ab} be a functional of matter data (the metric only enters index raising/lowering and normalization) and must not contain independent metric degrees of freedom.

Remark. The Lovelock theorem's literal premise "in 4 dimensions" means it cannot be used to derive $D = 4$; dimension must be argued separately (§6).

3.7 Locality Bifurcation and Excluded Routes

Lemma 3.12. The discrete Dirichlet energy is local; the effective resistance distance $R_{ij} = (e_i - e_j)^{\mathsf{T}} L^{+} (e_i - e_j)$ is nonlocal.

Lemma 3.13 (Limitations of the algebraic assembly route). The curvature of an equilateral complex is locked by combinatorial structure: the deficit angles around edge k are $+148.41$ř$, +77.88$ř$, +7.36$ř$, -63.17$ř $(k = 3, 4, 5, 6)$, with only $k = 3, 4, 5$ giving positive deficit. Algebraic assembly can therefore only land on an equilateral, rigid subset and cannot approximate a general metric.

Excluded route	Basis
Algebraic assembly $(K \to h)$	Unique fixed point = uniform ⇒ equilateralization ⇒ one-parameter rigidity
Effective resistance as geometry	Nonlocal; external edges break local isotropy
Homogeneity degree for dimension selection	Homogeneity degree is 1 for every d; does not select dimension

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4 Metric Construction and the Lovelock Lever

4.1 Common Weight Formula

All three metric routes use the same weight formula

$$w_{ij} = \phi_i \phi_j A_{ij}, \qquad h = \text{graph Laplacian with weight } w \text{ restricted to } H_Q, \tag{4.1}$$

differing only in the source of ϕ.

Route 1 (closed-walk counting, $k \to \infty$). The total traversal count of edge (i, j) by length-k closed walks is

$$N_{ij}^{(k)} = k\, A_{ij}(A^{k-1})_{ij}. \tag{4.2}$$

After normalization, $w_{ij} \propto \phi_i \phi_j A_{ij}$, with ϕ the Perron vector. Zero free parameters. But w depends on the Perron eigenvector—a global quantity of the entire graph—so (L') (g is locally determined by data) fails.

Route 2 (finite truncation $k = n$). Truncating (4.2) at scale k:

$$w_{ij}^{(k)} = \sum_{m=2}^{k} m\, A_{ij}(A^{m-1})_{ij}. \tag{4.3}$$

Locality holds exactly when $k \leq$ graph diameter (the effect of distal perturbations on central edge weight ratios is exactly zero). $k > n$ counts unrealizable closed words; $k < n$ misses realizable ones, so $k = n$ is forced.

Route 3 (locally given ϕ). If ϕ is given locally, (L') holds exactly, and all three premises are satisfied.

Remark (Scope of the (O) and (C) conditions). The second-order (O) and conservation (C) checks in Routes 1–3 strictly verify that the graph Laplacian L is a second-order, charge-conserving operator on scalar functions ($\ker L = \mathrm{span}\{\mathbf{1}\}$), which guarantees that the emergent spatial metric h is well-defined. They do not automatically imply that the field tensor $E_{ab}[g]$ is divergence-free under full four-dimensional spacetime diffeomorphisms; four-dimensional covariance $\nabla^a E_{ab} = 0$ remains an essential macroscopic symmetry postulate registered in §7.1 (Item 8).

4.2 Degrees of Freedom and Parameterization

The number of spatial metric components is

$$\dim \mathrm{Sym}^2 H_Q = \binom{m}{2} = \frac{(m-1)m}{2}. \tag{4.4}$$

Key distinction. (4.4) counts spatial metric h components ($m = 4$, 3D space: 6). Our construction $g = d\tau^2 - h$ (ADM $N = 1, \beta = 0$) fixes $g_{00} = 1$, $g_{0i} = 0$; graph edge weights provide only the 6 components of h, never the 10 components of the 4D spacetime metric g_{ab}.

Parameterization is not bijective. The weight formula (4.1) uses a vertex field $\phi = (\phi_1, \ldots, \phi_m)$ to give edge weights $w_{ij} = \phi_i \phi_j A_{ij}$. ϕ has only m free parameters, while $\dim \mathrm{Sym}^2 H_Q = \binom{m}{2}$; on K_m, the $\binom{m}{2}$ edge weights from m vertex values always satisfy the cross-product constraint $w_{12}w_{34} = w_{13}w_{24} = w_{14}w_{23} = \phi_1\phi_2\phi_3\phi_4$. It therefore spans only a $\leq m$-dimensional submanifold of $\mathrm{Sym}^2 H_Q$, not a bijection. The injectivity of $w \mapsto h$ (for independent weights) holds, but the ϕ parameterization of (4.1) is a local parameterization on a submanifold, not proven to cover the full space.

Graviton degrees of freedom. In general 4D GR: 10 metric components -4 coordinate gauge -4 initial-value constraints $= 2$ physical propagating degrees of freedom, i.e. $\frac{D(D-3)}{2} = 2$.

4.3 Refinement Convergence and Nonlocality of the Limit

The edge-weight ratio converges with truncation k, but extremely slowly, and the limit is the Perron form $P_1 = vv^{\mathsf{T}}$, globally nonlocal. The deviation decay length is

$$\tau \approx \frac{2N^2}{3\pi^2}. \tag{4.5}$$

Conclusion. The effective metric has a limit under refinement; the limit is the Perron form, nonlocal $\Rightarrow$ there exists no UV-fixed local metric.

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5 Continuum Limit and Causal Structure

5.1 5.1 Γ-Convergence

Lovelock requires Γ-convergence (local structure convergence), not a UV fixed point (self-similarity) [7]. A standard fact of homogenization theory: non-self-similar coefficient families still Γ-converge. With fixed step count $k = 8$, the profile maximum difference between adjacent N converges at approximately $O(a^2)$.

Boundary. Γ-convergence is not proven (no compactness, no Γ-liminf); only convergence and order are measured, and only on a 1D chain with a single profile.

5.2 Causal Structure: Support Cone and Cone Edge

Finite causal velocity requires saturation (per-point capacity), not current relaxation. Out-of-cone signals are exponentially small but nonzero pulled fronts [8].

The support cone is always exact: nearest-neighbor transport expands at most one lattice site per step, $\operatorname{supp} \rho_n \subseteq [-n, n]$, independent of B.

Cone-edge visibility is derived analytically: on the $e^{-\mu x}$ mode, the one-step mapping characteristic equation is

$$\Lambda^2 = B \cosh^2 \mu \implies \lambda(\mu) = \tfrac{1}{2} \log B + \log \cosh \mu. \tag{5.1}$$

The cone-edge density decay rate is

$$\varepsilon(B) = \tfrac{1}{2} \log\Big(\tfrac{4}{B}\Big). \tag{5.2}$$

Visible cone edge $\iff B = 4$. $B = 4$ is upgraded from a model parameter to a self-consistency condition.

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6 Dimension Account

This is the weakest link of the entire chain.

6.1 The Core Postulates Do Not Constrain Dimension (No-Go)

Proposition 6.1 (Classification-level no-go). Let $\mathcal{A}$ be the substrate core postulates. Then for every integer $m \geq 2$, there exists a model $\mathcal{M}_m$ of $\mathcal{A}$ with channel-set cardinality $|C| = m$.

Proof. Take $C = \{0, \dots, m-1\}$, $E = \{\{i, i+1\} : i \in \mathbb{Z}_m\}$ (cycle graph C_m). Verify postulate by postulate: C_m is finite and connected; every move has an inverse; all $2m$ directed moves are taken with equal weight; step index progresses irreversibly; closed state $x = 0$ exists. Numerical verification ($m = 2, \dots, 12$): $\operatorname{rank} B = m - 1$, $\ker L = \operatorname{span}\{\mathbf{1}\}$, $L|_{H_Q} \succ 0$. C_m is edge-transitive, so the no-preference $\Rightarrow$ uniform conductance premise also holds. ■

$$\implies \text{any statement depending only on } D \text{ cannot be derived from the substrate core postulates.} \tag{6.1}$$

6.2 Excluding Low Dimensions

D	Conclusion	Basis
2	No gravitational dynamics	$R_{ab} = \frac{1}{2} R g_{ab} \implies G_{ab} \equiv 0$
3	No graviton	Weyl tensor vanishes identically; curvature fully determined by local matter
4	Propagating graviton exists	see §6.3

6.3 6.3 Uniqueness of $D = 4$ from Graviton Degrees of Freedom

The D-dimensional graviton degrees of freedom are $\frac{D(D-3)}{2}$:

D	2	3	4	5	6	7	
$\frac{D(D-3)}{2}$		-1	0	**2**	5	9	14

$$\frac{D(D-3)}{2} = 2 \iff D^2 - 3D - 4 = 0 \iff D \in \{-1, 4\} \implies D = 4. \tag{6.2}$$

Screening chain. $\{2, 3, \dots\} \xrightarrow{D(D-3)/2>0} \{4, 5, \dots\} \xrightarrow{\text{minimality}} 4.$

Boundary. Minimality, even-dimension reading, and sign–polarization identification are not consequences of the substrate postulates.

6.4 Candidate Constraints as Synonymous Reformulations

Of 15 candidate constraints tested item by item, items 1–10 all have survival set $\{4\}$; they are all synonymous reformulations of $(Z_2)'$ and do not constitute independent evidence.

Among them, item 7, $b_1(K_m) = m - 1$, is examined in detail below.

6.5 Representation-Theoretic Status of the Cycle–Displacement Duality

As an **oriented** cycle space, $Z_1(K_4)$ decomposes as the standard S_4-module $(2,1,1) \cong (3,1) \otimes \mathrm{sgn}$, which is the representation of axial vectors (pseudovectors, flux, magnetic field). The displacement space is $H_Q = (3,1)$, the standard polar-vector representation. The two are distinct S_4-modules (they differ by the sign representation). **However, in three-dimensional geometric realization** (K_4 embedded as a regular tetrahedron in $\mathbb{R}^3$), the Hodge star operator

$$\star : \Lambda^2\mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

establishes a canonical duality between the axial cycle space and the polar displacement space. This is not a numerical coincidence: it is the group-theoretic counterpart of §6.8(ii) "antisymmetric composition closure"—the identity $\Lambda^2\mathbb{R}^3 \cong \mathbb{R}^3 \otimes \mathrm{sgn}$ is precisely the Hodge duality between closed 2-forms (loops, flux) and 1-forms (displacements) in three dimensions [9].

6.6 Dimension-Blind No-Go

All four metric-side structures $u = h/\sqrt{n}$, $\mathrm{H}(w)$, $\min(k, n-k)$, $T(M,n)$ do not reference the adjacency matrix A and take the same value on any graph.

$$\Longrightarrow \textbf{ the metric side cannot exclude low dimensions (e.g., } d = 1). \tag{6.3}$$

6.7 The Closure–Motion Correspondence Principle

The core postulates do not constrain dimension (Proposition 6.1). What fixes the dimension is this correspondence:

> **Closure–motion correspondence [named input].** The axiom, read at the counting level: $\dim\ker B = \dim\mathrm{im}\, B$ (degrees of freedom of closed loops = degrees of freedom of motion).

Here B is the incidence matrix of compensatory moves: $\mathrm{im}\, B = H_Q$ is the zero-sum state, and $\ker B$ is the closed-loop space.

$$\text{dictionary}: \quad d = m - 1, \quad \text{uniquely fixed by } \binom{m}{2} - (m-1) = \binom{m-1}{2}; \tag{1}$$

$$\text{reading}: \quad \binom{m}{2} - (m-1) = m - 1 \iff m = 4 \quad \text{(unique)}; \tag{2}$$

$$\text{no preference (edge-transitive)}: \quad \deg \text{ equal} \Longrightarrow t\,m = 4(m-1) \iff (\deg, m) = (3,4). \tag{3}$$

Machine check: on K_m, $\dim\ker B = \binom{m}{2} - (m-1)$ and $\dim\operatorname{im} B = m-1$ are equal only when $m = 4$ ($m = 2$ gives $0 \neq 1$, $m = 3$ gives $1 \neq 2$, $m \geq 5$ always gives $\dim\ker B > \dim\operatorname{im} B$).

Together the three give: **spatial dimension** $d = 3$**, channel count** $m = 4$**,** deg $= 3$**, spacetime dimension** $D_{\text{st}} = d + 1 = 4$.

$D_{\text{st}} = 4$ follows from two independent routes: the substrate-side premise is the closure–motion correspondence, th
(6.5)

Why the correspondence cannot be a consequence of the postulates. The substrate gives $\Gamma = (C, E)$ together with "motion"; the postulates only constrain *how* to move—"full branching" means every *allowed* motion is instantiated, and does not constrain which pairs have allowed moves. Any condition of the form "Γ must be complete/self-dual" is a choice of Γ, not a consequence of the postulates. Counterexample: cycle graph C_m ($m \geq 4$) satisfies the postulates and edge-transitivity, but $\dim\ker B = 1 \neq m - 1$.

6.8 Equivalent Formulations of the Correspondence Principle

The counting content $\binom{d}{2} = d$ has three equivalent formulations:

(i) Pairing uniqueness. Each edge has exactly one opposite edge $\iff \binom{m-2}{2} = 1 \iff m = 4$.

(ii) Antisymmetric composition closure. $\dim\Lambda^2 V = \dim V \iff D = 3$: the "relative relation" of two directions is again a direction (cross-product structure). I.e. $\binom{D}{2} = D \iff D \in \{0, 3\}$, with $D = 3$ the unique nontrivial solution.

(iii) A_3 **self-duality.** The 6 edge directions $\pm(e_i - e_j)$ of K_4 are exactly the 6 positive roots of A_3; $\mathrm{Aut}(K_4) = S_4$ is the Weyl group of A_3. Among A_n, only A_3 satisfies "number of relations = dimension": $\binom{n+1}{2} - n = n \iff n = 3$. This is precisely the accidental isomorphism $A_3 \cong D_3$ ($\mathfrak{so}(3) \cong \mathbb{R}^3$ cross-product isomorphism).

$\implies \Gamma = K_4$: 4 vertices, deg $= 3$, 6 edges, 3 opposite-edge pairs; angle $= \arccos(-1/3) = 109.4712$ř; $\mathrm{Aut}(K_4) =$
(6.6)

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7 Axiomatic Budget

7.1 What the Lovelock Lever Presupposes

#	Presupposition	Nature
1	$D = 4$ (literal hard-coding)	Cannot select dimension via this lever
2	Metric exists and is smooth	Input
3	Metric-volume consistency $\mu = \mathrm{vol}_h$	Sole additional input of (3.1)
4	Second-order (O)	Excludes higher-order terms

#	Presupposition	Nature
5	Conserved source (C')	(C) as-is is vacuous
6	Source = relativistic matter	Additional claim
7	Metric nondegenerate and Lorentzian	Input
8	4D diffeomorphism covariance $\nabla^a E_{ab} = 0$	Input: the preferred foliation of $g = d\tau^2 - h$ gives only 3D spatial diffeomorphism invariance

7.2 Absence of Intrinsic Scales

The dimensionful degrees of freedom in the geometric sector are exactly 1. (7.1)

Evidence. The layer coefficient (edge $(0,1)$) $\binom{m-1}{m/2}$ is an integer ($m = 2, 4, 6, 8$ gives $1, 3, 10, 35$). All raw metric-side data are integers $\Rightarrow$ they cannot carry length.

Unit-transformation invariance. Under $W \to sW$ (s spanning 8 orders of magnitude), normalized eigenvalue deviation is 1.89×10^{-15}, edge-weight ratio deviation is 0.

On G, Λ. Dimensional constants are units, not parameters [10]. "Deriving the numerical values of $G, \Lambda, \hbar$" is therefore not a well-posed problem.

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8 Main Results and Boundaries

8.1 Main Result

substrate (zero-sum constraint) + (inputs: embedding / source class / scale)

$$\Longrightarrow (L) \wedge (O) \wedge (C') \tag{8.1}$$

$$\Longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab} \quad (\text{in } D = 4).$$

the shape of the field equations is uniquely forced; the metric and dimension are inputs; the source is a no-go. (8.2)

8.2 Layering of "Dimension Is an Input"

"Input" here refers specifically to the Lovelock layer: the shape theorem $E_{ab} = \alpha G_{ab} + \beta g_{ab}$ takes D as an argument, and Lovelock itself does not select dimension. But the value $D = 4$ is not a bare free input; it is given by the readout chain: peak condition locks $L = 4$ (Gate 1) + closure-equality $\dim\ker B = \dim\mathrm{im}\, B$ locks K_4/A_3. This readout chain is half-locked: the peak condition hard-presses to $L = 4$; the step "$L = 4$ is exactly the physical spacetime dimension" is only count-level equality—$Z_1(K_4) \cong (2,1,1) \cong (3,1) \otimes \mathrm{sgn}$ and $H_Q = (3,1)$ are distinct S_4-modules differing by a parity sign, but in three-dimensional geometry form an axial–polar vector duality via the Hodge star.

8.3 Boundaries That Must Be Stated

1. **Division of labor of Lovelock**: it only answers "what must the field equations look like," not "why does geometry have dynamics."
2. **Local K_4 fiber vs. macroscopic grid are two structures.** $\dim H_Q = m-1 = 3$ holds only for the *single-point* local K_4 fiber; on the macroscopic N-vertex grid, $\dim H_Q = N-1 \neq 3$. The construction bridge "how local K_4 fibers/cells tile a macroscopic 3D manifold grid" is missing; it is an open item.
3. $L = 4$ **uses Gate 1 only.** The peak condition locks $L = 4$; this $k = 2$ partition does not pass the contextuality gate or the density gate, so it serves only as dimension readout, not as a quantumness criterion.

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9 Open Problem

Continuum limit (Γ-convergence) is unproven: no compactness, no Γ-liminf; only on a 1D chain with a single profile. This is analytic (compactness / variational limit) and requires independent work in homogenization theory.

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10 Conclusion

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\begin{aligned}
&\text{substrate (zero-sum constraint)}\\
&\quad\Longrightarrow \dim H_Q = m-1;\ \operatorname{im} B = H_Q;\ \text{discrete Dirichlet energy;}\\
&\quad\Longrightarrow (\text{continuum + metric-volume consistency}) \text{ spatial metric } h;\\
&\quad\Longrightarrow (\text{reversible/irreversible splitting}) \text{ signature } (1, m-1);\\
&\quad\Longrightarrow (\text{locality} \wedge \text{second-order} \wedge \text{conserved source})\\
&\quad\Longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab} \quad (\text{in } D = 4).
\end{aligned}
\tag{10.1}
$$

Shape is unconditional; entry is conditional: $(L) \wedge (O) \wedge (C') \Rightarrow$ Einstein form is unconditional (Lovelock); dimension, metric, and source are three entries, all non-consequences of the postulates.

Current progress is **conditional recovery**, not **unconditional derivation**. To reach *derivation*, the same ledgers are missing: the premise of dimension, the origin of the metric, and the price tag of the source.

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